



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
UNIT TEST
18-07-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PHYSICS

SEC-1 : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

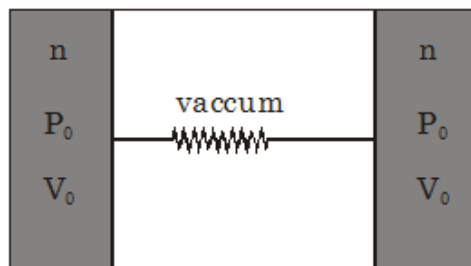
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Consider a vernier caliper. Least count of main scale is 1mm. There are 20 equispaced marks on vernier scale. When nothing is put between the jaws, zero of main scale lies between zero and first division of vernier scale and 15th division of vernier exactly coincide with one of the main scale division. When zero of vernier is matched with zero of main scale, 20th division of vernier exactly coincides with 25th division of main scale. To measure the side of a cuboid, it is kept between the jaws of vernier caliper, it is now observed that 18th division of vernier exactly coincides with 32nd division of main scale. Choose the correct option(s).

- (A) Least count of vernier scale is 0.25 mm (B) Zero error is -0.25 mm
 (C) Side of cuboid is 10.25 mm (D) Side of cuboid is 9.25 mm

Space for Rough Work

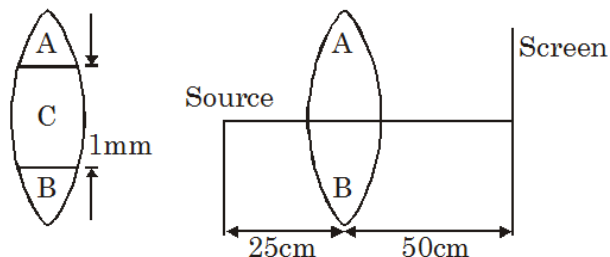
2. In the arrangement shown in the figure, the cylinder is insulated from the surrounding. On both sides, same diatomic gas is trapped by two insulating massless pistons with the help of an ideal spring. The natural length of the spring is equal to the length of the cylinder. Initial state of the gases are as shown in the figure.



- (A) The value of energy stored in the spring is $P_0 V_0$.
- (B) The value of energy stored in the spring is $2P_0 V_0$.
- (C) Now the gases are heated slowly, such that their temperature becomes three times of their initial temperature. The total heat given to the system is $12P_0 V_0$.
- (D) Now the gases are heated slowly, such that their temperature becomes three times of their initial temperature. The total heat given to the system is $6P_0 V_0$.
3. The height of a building may be estimated by measuring the horizontal distance to a point on the ground and the angle from this point to the top of the building. Assuming these measurements are $L = 100\text{ft} \pm 0.5\text{ ft}$ and $\theta = 30^\circ \pm 0.2\text{ degrees}$. Estimate the height h of the building and the uncertainty in the estimate. For the same building height and measurement uncertainties, determine the angle $\theta_{\min}$ (and the corresponding distance $L_{\min}$ from the building) at which measurements should be made to minimize the uncertainty in estimated height.
- (A) $\theta_{\min} = 45^\circ$ (B) error in h is 1.3 %
- (C) $h = 57.7 \pm 0.2\text{ ft}$ (D) $L_{\min} = 57.7\text{ ft}$

Space for Rough Work

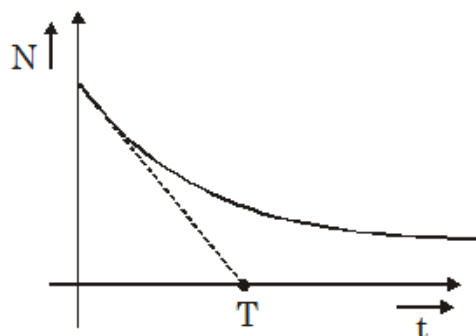
4. A convex lens of focal length 50 cm is cut along the diameter into two identical halves A and B and in the process a layer C of the lens thickness 1 mm is lost. Then the two halves A and B are put together to form a composite lens. Now in front of this composite lens a source of light emitting wavelength $\lambda = 6000 \text{ \AA}$ is placed at a distance of 25 cm as shown in the figure. Behind the lens there is a screen at a distance 50 cm from it. Choose the correct option(s):



- (A) Two image will be formed by lens which will result in interference pattern on the screen.
 (B) One image will be formed by lens which will produce diffraction pattern on the screen
 (C) Fringe width of the interference pattern obtained on the screen is 0.6 mm.
 (D) Fringe width of the central maxima on the screen will be 1.2 mm.

Space for Rough Work

5. A radioactive sample undergoes decay as per the following graph. Dotted line represents the slope of the curve at $t = 0$. If this dotted line intersects the time axis at $t = T$, choose the correct alternative (Given $t_{1/2}$ = half life, t_{avg} = mean life)



- (A) $t_{1/2} > T$

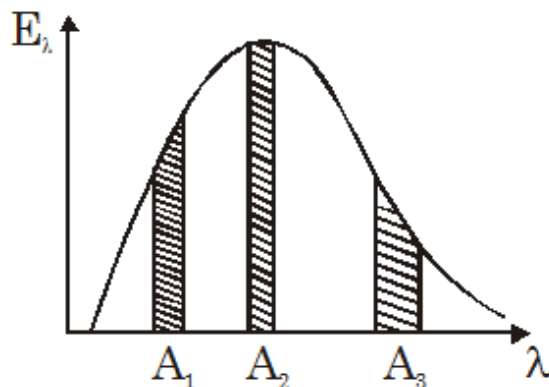
(B) $t_{\text{avg}} = T$

(C) $t_{1/2} = T$

(D) $t_{\text{avg}} > T$

Space for Rough Work

6. Three separate segments of equal area A_1 , A_2 and A_3 are shown in the energy distribution curve of a blackbody radiation. If n_1 , n_2 and n_3 are number of photons emitted per unit time corresponding to each area segment respectively then :



- (A) Energy emitted by the body per unit time corresponding to these wavelengths are all equal.
 (B) Energy emitted by the body per unit time corresponding to these wavelengths are in order $E_1 > E_2 > E_3$.
 (C) $n_1 > n_2 > n_3$
 (D) $n_1 < n_2 < n_3$

Space for Rough Work

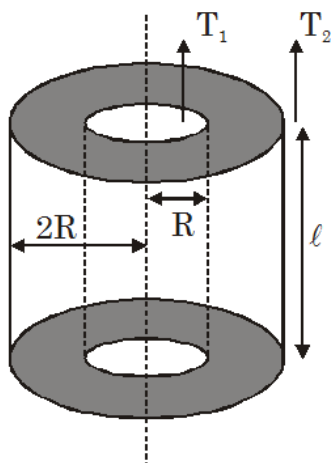
SEC-2 : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
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- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

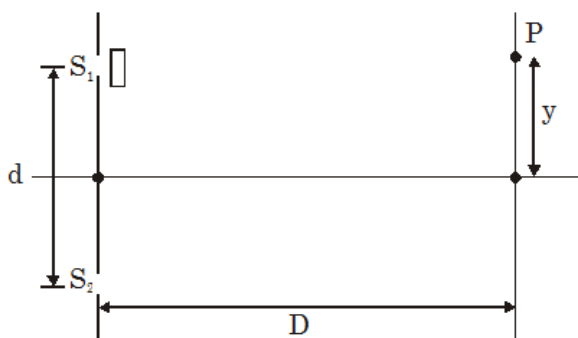
Zero Marks : 0 In all other cases.

1. Inner surface of a cylindrical shell of length ℓ and of material of thermal conductivity k is kept at constant temperature T_1 and outer surface of the cylinder is kept at constant temperature T_2 such that ($T_1 > T_2$) as shown in figure. Heat flows from inner surface to outer surface radially outward. Inner and outer radii of the shell are R and $2R$ respectively. Due to lack of space this cylinder has to be replaced by a smaller cylinder of length $\frac{\ell}{2}$ inner and outer radii $\frac{R}{4}$ and R respectively and thermal conductivity of material nk . If rate of radially outward heat flow remains same for same temperatures of inner and outer surface i.e. T_1 and T_2 , then find the value of n .

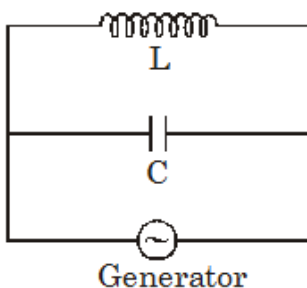


Space for Rough Work

2. A $50\ \Omega$ resistor is connected in series with a coil and a sinusoidal ac-source of effective voltage 50 V. If the effective voltage drop across the resistor is 20 V and across the coil is 40 V, Find the power developed in coil in watt?
3. Young's double slit experiment is conducted with white light ($4000\text{\AA} - 7000\text{\AA}$). As shown in the diagram, the slit separation $d = \left(\frac{1}{2}\right)$ mm and screen slit distance, $D = 1$ m. The upper slit is covered with a film of thickness, $t = 1\ \mu\text{m}$ having a refractive index, $\mu = \frac{3}{2}$. Find the number of wavelengths which will form a maxima at $y = 3\text{mm}$.



4. For the circuit shown in the figure, the current through the inductor is 0.6 A, while the current through the capacitor is 0.4 A. Find the current (in A) drawn from the generator.



Space for Rough Work

5. Consider the nuclear reaction ${}_{36}^{81}\text{Kr} + {}_{-1}^0\text{e} \rightarrow {}_{35}^{81}\text{Br} + {}_0^0\nu$. The mass of Kr is 80.9166 amu, mass of Br is 80.9163 amu, mass of electron is 0.00054858 amu and neutrino can be assumed to be massless. If it behaves like a photon, Br will recoil after the nuclear reaction. What is the recoil velocity of Br (in km/s) ? Take 1 amu = 931 MeV/ c^2 .
6. In Bainbridge mass spectrograph, ions are made to pass through a velocity selector and then through a magnetic field. The magnetic field in the velocity selector is the same as the magnetic field used to bend the trajectory of ions. The specific charge (q/m) of ions is to be calculated. The magnetic field is 0.100 ± 0.002 T, Electric field is 100 ± 1 N/C, and four measurements of radius of circular path were made. The readings were 0.100 mm, 0.110mm, 0.090 mm and 0.100 mm. What is the error in the value of specific charge in multiples of 10^7 C/kg?

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

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- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

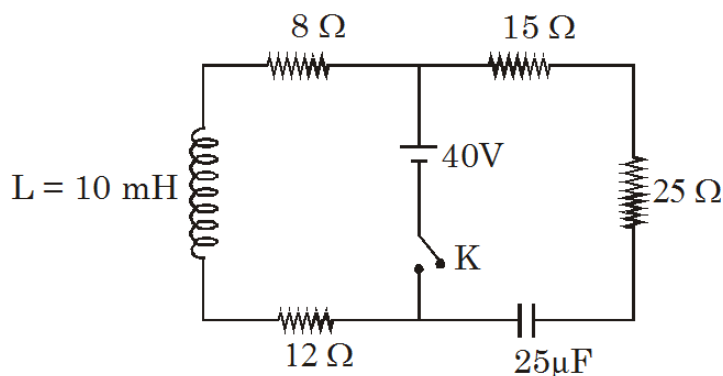
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

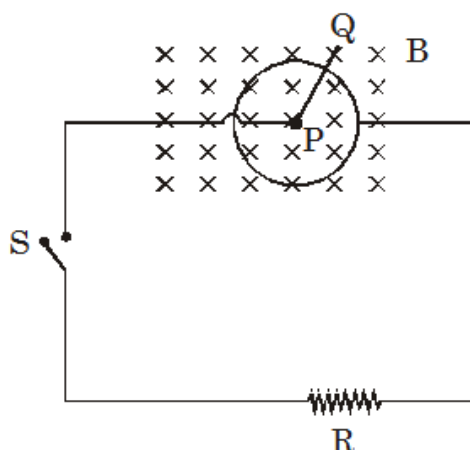
1. A ideal gas whose adiabatic exponent equals γ is expanded so that the amount of heat transferred to the gas equal to twice of decrease of its internal energy. The equation of the processes $T V^{\frac{\gamma-1}{K}} = \text{constant}$ (where T and V are absolute temperature and volume respectively). Find the value of K.
2. A coil of inductance $L = 5\text{H}$ and resistance $R = 55\Omega$ is connected in series to the mains alternating voltage of frequency $f = 50\text{ Hz}$ in series. What can be the non-zero capacitance of the capacitor (in μF) connected in series with the coil if the power dissipated has to remains unchanged? (take $\pi^2 = 10$)

Space for Rough Work

3. In the circuit the key (K) is closed at $t = 0$. Find the current in ampere through the key(K) at the instant $t = 10^{-3} \ln 2$ sec



4. A resistance $R = 1 \Omega$ connected to a conducting rod PQ that can slide on a conducting circular ring of radius 2m with center at P. A constant uniform magnetic field 1 Tesla exist as shown in figure. At $t = 0$, with 'S' is closed and simultaneously an external torque is applied on the rod so that it rotates with constant angular velocity 2 rad/sec. The magnitude of power delivered by external agent is 2 K Watt. Find the value of K. (Neglect the gravity, friction and self inductance of circular loop)



Space for Rough Work

5. A photoelectron was emitted normally from positive plate of a charge parallel plate capacitor. The voltage across the capacitor was 1V. The wavelength of incident light = 310 nm and work function was 2.2 eV. $m_e = 9 \times 10^{-31}$ kg, $hC = 1240$ eV nm. The electron reaches the other plate in 3 nsec what is distance between the plates (in mm)
6. A quantity x is defined as $x = \frac{a^2 - b^2}{\sqrt{c + d}}$. Value of a, b, c and d are reported as $a = 3.00 \pm 0.01$, $b = 5.00 \pm 0.01$, $c = 6.00 \pm 0.24$ and $d = 10.0 \pm 0.4$. Percentage error in x will be.

Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 24)

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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

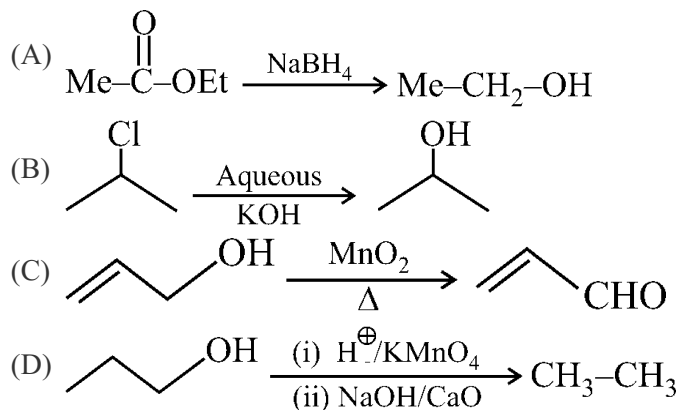
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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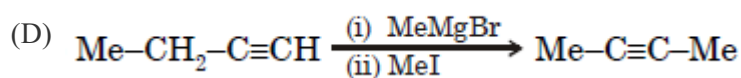
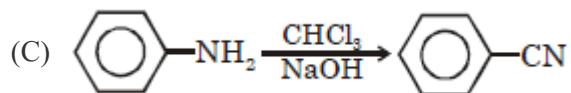
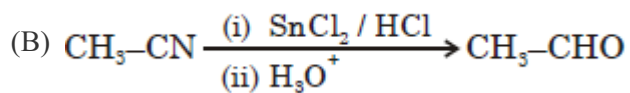
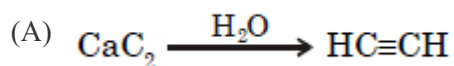
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1. Reaction correctly matched with their major product is/are :



Space for Rough Work

2. Which of the following reaction(s) is/are correctly matched with major product :



3. Compound "A" turns starch iodate into a deep blue solution in an acidic medium. This reagent A can also be obtained by boiling $[\text{AgSO}_3]^-$ solution
Which is/are CORRECT for "A"

- (A) It has two $p\pi$ - $d\pi$ bonds
- (B) It is more acidic compared to P_4O_6
- (C) It can act as a monodentate ligand
- (D) It can act as oxidizing as well as reducing agent.

Space for Rough Work

4. Which of the following reaction(s) will yield atleast a diatomic molecule that has H.O.M.O as gerade orbital
- (A) $[\text{Fe}(\text{H}_2\text{O}_5)\text{NO}]^{2+} + \text{H}_2\text{O} \xrightarrow{\text{Shaking}}$ (B) $\text{HO.SO}_2\text{NH}_2 + \text{HNO}_2 \rightarrow$
- (C) $\text{NH}_4\text{ClO}_4 \xrightarrow{\Delta}$ (D) $\text{NH}_4\text{NO}_2 \xrightarrow{\Delta}$
5. For $\text{MX}_2(\text{s})$ lattice enthalpy is 800kJ mol^{-1} and enthalpy of solution is 50kJ mol^{-1} . If hydration energies of M^{2+} and X^- ions are in the ratio 3 : 1, then correct options are :
- (A) Magnitude of hydration energy of M^{2+} is 300 kJ/mol
- (B) Magnitude of hydration energy of M^{2+} is 450 kJ/mol
- (C) Magnitude of hydration energy of X^- is 300 kJ/mol
- (D) Magnitude of hydration energy of X^- is 150 kJ/mol
6. An ideal monoatomic gas undergoes a process in which $P = 3V^3$. Choose the correct statements
- (A) If temperature of the gas decreases, then pressure must also decrease .
- (B) If volume increases, then temperature will also increase
- (C) For expansion, heat will have to be supplied to the gas
- (D) If temperature increases then work is positive

Space for Rough Work

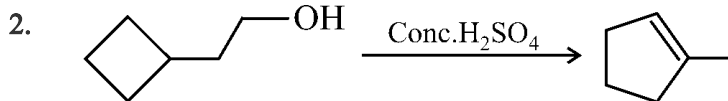
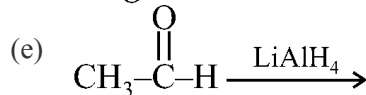
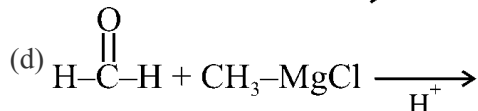
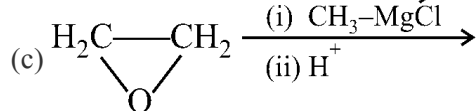
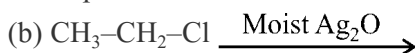
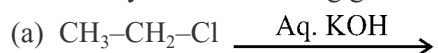
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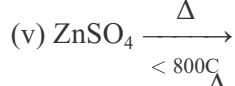
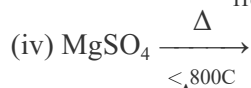
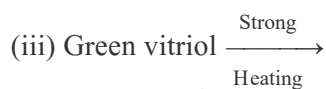
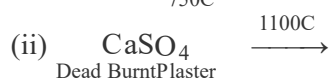
1. How many of the following give 1° alcohol as a product :



Total number of 1,2-shift involve in above reaction :

Space for Rough Work

3. Number of reactions given below that can produce SO_3



4. 0.006 M acetate solution with excess of neutral ferric chloride solution gives deep red color due to formation of a complex ion. The complex ion dissociates on heating to give a Brown red precipitate "A". Neglecting H^+ from any weak acid formed, the pH of final solution is
(Take $\log_{10} 3 = 0.47$)

Space for Rough Work

5. In the titration of a solution of a weak acid HX with NaOH, the pH is 5.8 after 10 ml of NaOH solution has been added and 6.40 after 20 ml of NaOH has been added.

If the ionization constant of HX is written as $a \times 10^{-b}$.

Find the value of $a + b$.

(take $\log 2 = 0.3$)

6. The reaction $2A + B \rightarrow C + D$ and follows the rate law $-\frac{d[A]}{dt} = K[A]^2[B]$.

The sum of values of x and y in the following data:

Set	$[A_0] \times 10^6 \text{ M}$	$[B_0] \times 10^6 \text{ M}$	half-life (s)
1	3×10^5	40	62.5
2	3×10^5	60	x
3	5	3×10^5	625
4	10	3×10^5	y

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

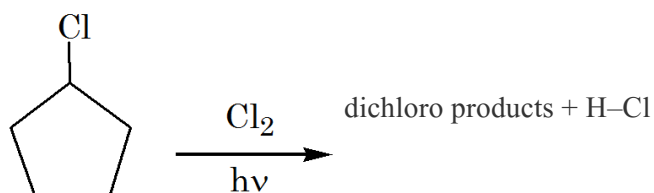
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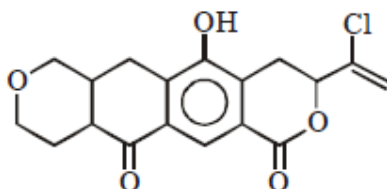
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1. Identify total number of theoretically possible dichloro products formed in following reaction -



2. When given compound react with excess of RMgX, then find the number of moles of G.R.(RMgX)



3. Number of species that can be used to distinguish AgCl from AgBr and AgI among the following HNO_3 , $Na_2S_2O_3$, NaCN, KCN, Na_3AsO_3 , NH_3 (solution)

Space for Rough Work

4.

Observation	Inference	
	Gas/vapour evolved	Possible Anion
A colourless gas with pungent smell, which gives dense white fumes when a rod dipped in ammonium hydroxide is brought near the mouth of the test tube.	A	G
Reddish brown gas with a pungent odour is evolved. Intensity of reddish gas increases on heating the reaction mixture after addition of solid MnO_2 to the reaction mixture. Solution also acquires red colour.	B	H
Violet vapours, which turn starch paper blue and a layer of violet sublimate is formed on the sides of the tube. Fumes become dense on adding MnO_2 to the reaction mixture.	C	I
Brown fumes evolve which become dense upon heating the reaction mixture after addition of copper turnings and the solution acquires blue colour.	D	J
Colourless, odourless gas is evolved which turns lime water milky and the gas coming out of lime water burns with a blue flame, if ignited.	E & F	K

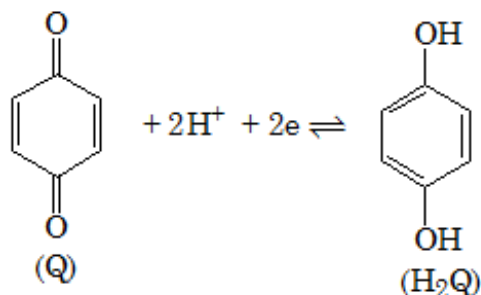
 If x = no. of paramagnetic species among A, B, C, D, E & F

 y = no. of species among G, H, I, J, K whose salt with Ag is soluble

 then $(x - y)$ is

Space for Rough Work

5. Quinhydrone is a sparingly soluble one-to-one addition compound formed from hydroquinone and quinone. When solid quinhydrone is dissolved in an aqueous medium, equal concentrations of hydroquinone and quinone result. If a platinum wire is dipped into the solution, the potential of the electrode is governed by the reversible reaction :



Where Q denotes quinone and H_2Q denotes hydroquinone. The E° value of this half cell reaction is $+0.46\text{V}$ with respect to the saturated calomel reference electrode (SCE). Because the potential of the quinone-hydroquinone half reaction depends on the concentration of hydrogen ion, it is possible to use this system to measure pH. A sample solution of unknown pH was saturated with quinhydrone and a platinum electrode was dipped into it. If the potential of the such electrode was found to be $+0.22\text{V}$ with respect to SCE, what was the PH of the sample solution?

[Given : $\frac{2.303 RT}{F} = 0.06$]

6. Energy is supplied to a gaseous mixture of H and He^+ in such a way that H atoms & He^+ ions are excited to their first excited state. Subsequently by collision between excited H atoms & He^+ ions, all excitation energy of H atoms is transferred to He^+ ions so that principal quantam no. of finally excited He^+ ion becomes x. Find x.

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Identify the correct statement(s)
(Where 'c' is constant of integration)

$$(A) \int \frac{(x^5 - x)^{1/5}}{x^6} dx = \frac{5}{24} \left(1 - \frac{1}{x^4} \right)^{6/5} + c$$

$$(B) \int \frac{(x^5 - x)^{1/5}}{x^6} dx = -\frac{5}{24} \left(1 - \frac{1}{x^4} \right)^{6/5} + c$$

$$(C) \int (\tan^3 x + \tan^2 x + \tan x + 1) dx = \frac{(\tan x + 1)^2}{2} + c$$

$$(D) \int (\tan^3 x + \tan^2 x + \tan x + 1) dx = \frac{1}{2} \sec^2 x + \tan x + c$$

Space for Rough Work

2. Consider a square matrix A . Identify the correct statement(s) -
- (A) If A is nilpotent matrix of index 2 then $A(I + A)^n = A \forall n \in \mathbb{N}$
- (B) If A is nilpotent matrix of index 2 then $A(I - A)^n = A \forall n \in \mathbb{N}$
- (C) If A is idempotent matrix, then $(I + A)^n = I + (2^n - 1)A \forall n \in \mathbb{N}$
- (D) If A is idempotent matrix, then $(I + A)^n = I - (2^n - 1)A \forall n \in \mathbb{N}$
3. Let graph of $f(x)$ passes through $(0, 2)$ and tangent at any point (x, y) of $f(x)$ intersects x-axis at $(x + 2, 0)$, then -
- (A) $\lim_{x \rightarrow \infty} f(x) = 0$
- (B) $\lim_{x \rightarrow \infty} f(x) = 2$
- (C) equation of normal at $(0, 2)$ is $x - y + 2 = 0$
- (D) equation of normal at $(0, 2)$ is $x + y - 2 = 0$

Space for Rough Work

4. F is the family of curves in the xy-plane, given by $y = x + ce^{-x}$, where $c \in \mathbb{R}$ is an arbitrary constant. The orthogonal trajectories of the curves of family F form the family F'. Then which of the following statements are true for F'.

- (A) $xe^y = e^y(y - 2) + c$, is general solution for F'
 (B) $x = y - 2$ is a particular solution for F'
 (C) If F' passes through (4, 0), then F' is $x = (y - 2) + 6e^y$
 (D) $xe = e(y - 2) + c$, is general solution for F'

5. Let $A_n = \sum_{k=0}^{n-1} \frac{n}{k^2 + n^2 + (n+2)k + n + 1}$ and $B_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2}$ for $n = 1, 2, 3, \dots$, then which of the following is/are true ?

- (A) $A_n < B_n$ (B) $A_n > B_n$ (C) $A_n < \frac{\pi}{3}$ (D) $B_n > \frac{\pi}{9}$

6. If A and B are two matrices such that $AB = A$ and $BA = B$, then

- (A) $(A^3 - B^3)^5 = (A^5 - B^5)^3$ (B) $(A^4 - B^4)^3 = A^{12} - B^{12}$
 (C) $(A - B)$ is idempotent (D) $(A^{10} - B^{10})$ is nilpotent

Space for Rough Work

SEC-2 : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. The area of the region which consists of all the points satisfying the conditions $|x - y| + |x + y| \leq 8$, $xy \geq 2$, $x \geq 0$ and $y \geq 0$ is equal to - (take $\ln 2 = 0.69$)
2. The area in sq. unit bounded by the curves $y^2 = x^3$ and $|y| = 2x$ is given by-
3. The value of $\int_0^1 \sqrt[3]{2x^3 - 3x^2 - x + 1} \, dx$ is -

Space for Rough Work

4. Let $A = \begin{bmatrix} \frac{1}{2} & \frac{3}{4} \\ 1 & -\frac{1}{2} \end{bmatrix}$ then the value of $\text{tr}(A) + \text{tr}(A^2) + \text{tr}(A^3) + \dots + \text{tr}(A^{100})$, is

(Where $\text{tr}(A)$ denotes trace of matrix A)

5. Area of the region $x^6 - x^2 + y^2 \leq 0$ is
6. If $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{3x^2 + mx + n}{x^2 + 1}$ has range $[-4, 3)$ then $m^2 + n^2$ is

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. The number of integral values of k for which the equation $e^x = \frac{k}{x-3}$, has exactly two solutions is-
2. Let a derivable function f satisfy the rule $f(x+y) = f(x) + f(y) + \frac{xy}{2} \forall x, y \in \mathbb{R}$ and $f'(0) = 0$. If the area bounded by $y = f(x)$ and the line $y = ax + 9$, ($a \in \mathbb{R}$) is minimum then 'a' is equal to
3. Let $g : \left[\frac{1}{2}, 1\right] \rightarrow (0, \infty)$, $g(x) = e^{-2x}f(x)$, where $g(x)$ is non constant decreasing and differentiable function and $g\left(\frac{1}{2}\right) = \frac{1}{e}$. Then the value of $\left\lfloor \int_{\frac{1}{2}}^1 f(x) dx \right\rfloor$ is
[where $[x]$ denotes the greatest integer less than or equal to real number x]

Space for Rough Work

4. Given f is a differentiable function such that $1 \leq f(x) + f'(x) \leq 2 \forall x \in \mathbb{R}$ and $f(0) = 0$, then the value of $\left[\frac{f(1)}{2} \right]$ is equal to
(where $[x]$ denotes the greatest integer less than or equal to real number x)
5. Let $\int \frac{x^2 + 20}{(x \sin x + 5 \cos x)^2} dx = f(x) + \tan x + C$, $f(0) = 0$ where C denotes constant of integration. Then the value of $\pi + 5f(\pi)$ is equal to
6. If $\int \frac{(3x^2 + 2x)dx}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5} = \frac{1}{2} \tan^{-1}(g(x)) + C$ (where $g(0) = \frac{1}{2}$), then the value of $2g(1)$ is

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work